

Path

↳ Spin representation $\uparrow$ new way
↳ V_2 model "works" to compute

Adiabatic quantum computing AQC

D-Wave Canada / USA

↳ 5000 qubits.

not the same as in gate-based QC.

Objective AQC:

↳ find arg-min $H(\underline{\sigma})$.

$H(\underline{\sigma})$
HARD

$Q_0(\sigma)$
EASY

Easy: $H_0(\sigma) = \frac{1}{2} \sum_{n,m} \hat{\sigma}_n^z \hat{\sigma}_m^z$

Hard: $H(\underline{\sigma}) = -\frac{1}{2} \sum \underline{A_{nm}} \hat{\sigma}_n^x \hat{\sigma}_m^x$

all spins should be $\uparrow\uparrow$ or $\uparrow\downarrow$.
with the x-axis.
we don't know which.

$$\tilde{H}(\sigma, t) = \underbrace{H_0(\sigma)}_{t=0} (1-t) + \underbrace{H(\sigma)}_{t=1} \cdot t.$$

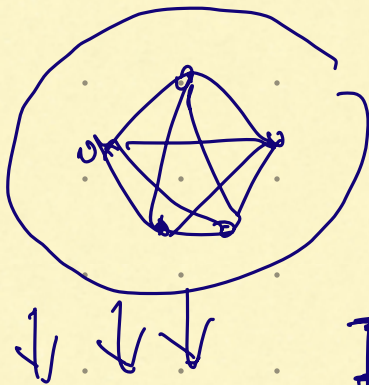
slow change = adiabatic

Quantum adiabatic theorem

if at $t=0$, $|\psi(t=0)\rangle \leftarrow$ ground state of $H_0(\sigma)$
 $\tilde{H}(\sigma, t=0)$

and t changes slowly

$|\psi(t=1)\rangle$ is also a ground state
 but $H(\sigma)$
 $\tilde{H}(\sigma, t=1)$



$$H_0(\sigma) = -\frac{1}{2} \sum_{m,n} \sigma_m \sigma_n$$

N^2 -fold

$N^2 \leftarrow$ degenerate
 $\leftarrow$ a lot N
 $\bullet \leftarrow$ ground state (unique).



Removed degeneracies.

$$\Delta p \cdot \Delta x \gtrsim \hbar$$

Heisenberg

This is not about measurements!

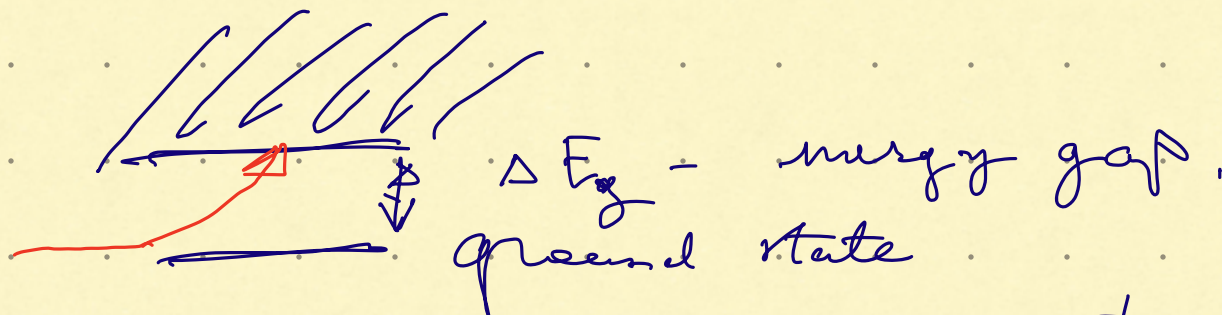
$$\Delta E \cdot \Delta t \gtrsim \hbar$$

$$\Delta t \sim 10 \text{ ns}$$

Direct classical correspondence

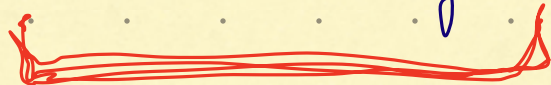
$$\Delta T$$

$$\Delta \omega \cdot \Delta t > \pi$$



$$\Delta t \lesssim \frac{\hbar}{\Delta E_g}$$

If $\Delta t \gg \frac{\hbar}{\Delta E_g}$



moving away from the ground state would break UR,

Is ΔE_g large enough?



$$\Delta E_g \sim 0$$

common for really hard problems.

HP - hard problem

↓
Find good schedule.